

# Characterisation of the `reco_far` events for detector 3

(June cosmic bench — `g_det3_wknd`)

MX17 cosmic-bench QA

July 5, 2026

## 1 Setup and definitions

For every clean M3 reference muon crossing the `det3` active area we ask where the detector’s reconstructed  $X+Y$  point lands relative to the M3 ray projection:

- `reco_near`: valid reco point,  $|r| \leq 5$  mm (this is the efficiency);
- `reco_far`: valid reco point, but  $|r| > 5$  mm;
- `spark`: event fires  $> 50$  strips (a full-detector discharge), pulled out *before* the reco/hit split so it does not contaminate the tail;
- `hit_no_reco` / `no_hit`: fired but no valid point / silent.

Run `mx17_det3_p2_det1_overnight_6-27-26 / long_run_p2`, the headline `det3` page. Numbers below reproduce the June overview PDF exactly.

| category                              | $N$          | % of active-area crossings |
|---------------------------------------|--------------|----------------------------|
| <code>reco_near</code> ( $\leq 5$ mm) | 41 743       | 78.8 %                     |
| <code>reco_far</code> ( $> 5$ mm)     | <b>7 271</b> | <b>13.7 %</b>              |
| <code>spark</code> ( $> 50$ strips)   | 2 600        | 4.9 %                      |
| <code>hit_no_reco</code>              | 85           | 0.2 %                      |
| <code>no_hit</code> (silent)          | 1 296        | 2.4 %                      |
| total crossings                       | 52 995       | 100 %                      |

Table 1: `det3` breakdown (sparks already removed). The 13.7 % `reco_far` tail is the subject of this note.

## 2 Result in one paragraph

The `reco_far` tail is *not* uniform and *not* a tracking artefact. It is two overlapping populations: (i) a **near-miss** shoulder just outside the 5 mm cut (41 % of `reco_far` sit at 5–10 mm), which is the ordinary resolution/angle tail; and (ii) a genuine **mis-reconstruction** tail (38 % beyond 20 mm, 23 % beyond 50 mm) that is driven by **elevated-multiplicity “sub-veto” discharge activity**, spatially **concentrated on one side of the detector (low  $X$  / left edge)**. The multi-ray / ray-mismatch hypothesis is cleanly ruled out (0 % of `reco_far` events have more than one M3 ray).

## 3 Where they are: a localised low- $X$ (left-edge) region

Figure 1 is the key plot. Both the muon crossings that go `reco_far` (panel a) and — more sharply — the detector positions where the bad reco point *lands* (panel c) pile up on the low- $X$  (left) side

of the chamber. The `reco_far` rate map (panel b) is  $\sim 10\%$  in the bulk but rises to 25–70% along the left edge and around the periphery (median 14%, 90<sup>th</sup> pct 25%). A wrong reco point preferentially reconstructs into the low- $X$  band regardless of where the muon actually crossed — the signature of a localised noisy / spark-prone detector region generating displaced clusters, not a global reconstruction bias.

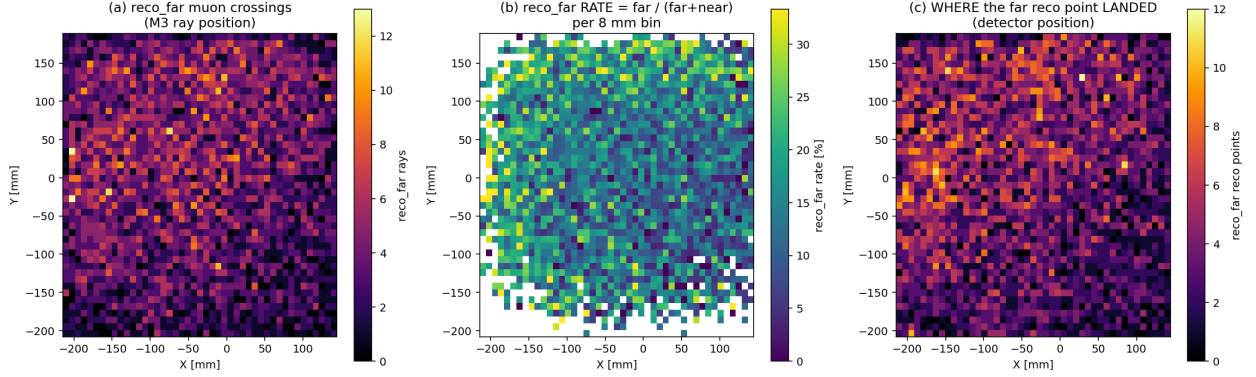


Figure 1: **Spatial distribution.** (a) M3 crossing positions of `reco_far` muons; (b) `reco_far` rate =  $\text{far}/(\text{far}+\text{near})$  per 8 mm bin (bins with  $\geq 15$  tracks); (c) detector positions where the far reco point landed. The low- $X$  / left-edge region dominates both the rate and the landing points.

## 4 What kind of events: elevated multiplicity, sub-veto discharges

`reco_far` events fire *more* strips than `reco_near` ones (median 20 vs 16; 90<sup>th</sup> pct 39 vs 23), with a population piling up against the 50-strip spark veto (Fig. 2a). These are “half-sparks”: the same discharge phenomenon as the vetoed sparks, but sitting in the 20–50 strip shoulder just below the cut, where they survive into the reco categories and throw spurious displaced clusters. The effect strengthens with  $|r|$ : the genuinely-wrong tail ( $> 20$  mm) has median multiplicity 26 versus 18 for the 5–10 mm near-misses. A long-duration cluster tail corroborates this — the mean  $X$  cluster duration is 3888 ns for `reco_far` versus 581 ns for `reco_near` (Fig. 2c), i.e. a subset of multi-pulse / afterpulsing clusters. Fitted cluster sizes are only mildly larger (median 9 vs 8 strips, panel b).

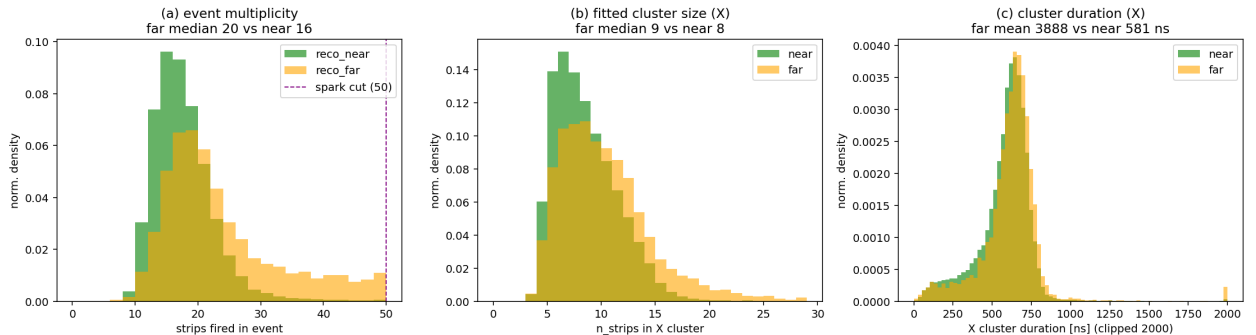


Figure 2: **Event character (far vs near, area-normalised).** (a) strips fired per event — `reco_far` extends toward the 50-strip spark cut; (b) fitted  $X$ -cluster size; (c)  $X$ -cluster duration — `reco_far` has a heavy long-duration (multi-pulse) tail (note the pile-up at the 2000 ns clip).

## 5 Which plane fails, and the multi-ray cross-check

No single readout plane is responsible: of the `reco_far` events, 35.6% have only the  $X$  plane displaced, 23.5% only  $Y$ , and 40.9% have *both* planes wrong (threshold  $5/\sqrt{2} \approx 3.5$  mm). The per-plane displacement map (Fig. 3b) shows the corresponding “cross-plus-cloud”: bright axis arms (one plane correct, one wrong = the near-miss shoulder) sitting on a diffuse both-wrong cloud (the genuine tail). The radial residual (Fig. 3a) falls smoothly from the 5 mm cut and flattens into a wide  $> 50$  mm plateau extending across the whole chamber — consistent with clusters landing at essentially random wrong positions.

Finally, the ray-mismatch hypothesis is excluded: 0.00% of `reco_far` (and of `reco_near`) events carry more than one accepted M3 ray after the  $\chi^2 < 20$  cut, so a “detector saw the other track” explanation does not apply to this run.

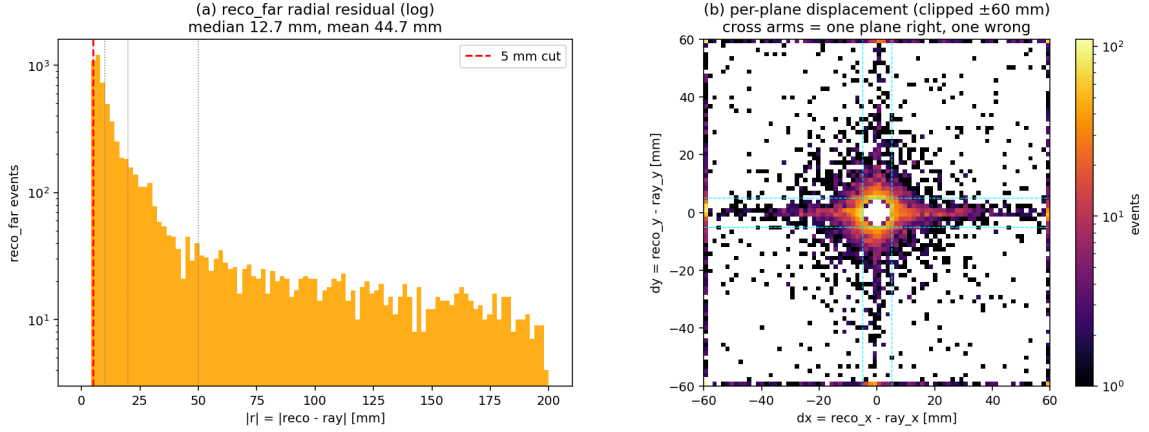


Figure 3: **Residual structure.** (a) `reco_far` radial residual (log  $y$ ): smooth fall-off plus a flat far plateau. (b) per-plane displacement  $(dx, dy)$ , clipped  $\pm 60$  mm; cyan lines mark  $\pm 5$  mm. The cross arms are one-plane-wrong events; the diffuse cloud is both-planes-wrong.

## 6 Interpretation and takeaways

- About 40% of the `reco_far` tail is a benign 5–10 mm near-miss shoulder — the resolution/angle tail spilling just past the 5 mm cut. Widening the cut to  $\sim 7$  mm, or the existing core- $\sigma$  metric, already accounts for it.
- The remaining, genuinely-wrong tail ( $\gtrsim 20$  mm,  $\sim 38\%$ ) is a **detector/HV effect**: sub-veto discharges (20–50 strips, long multi-pulse clusters) concentrated in the **low- $X$  / left-edge** region, producing displaced competing clusters that the micro-TPC fit sometimes selects.
- It is *not* a tracking or ray-matching problem (multi-ray = 0%) and *not* a single-plane readout fault (both planes contribute).
- **Suggested follow-ups:** (1) tighten the discharge veto toward  $\sim 30$ –40 strips (or add a cluster-duration / multi-pulse veto) and re-measure — most of the genuine tail should move into a **spark-like** category; (2) inspect the low- $X$  edge strips for HV/sparking or common-mode noise; (3) at the waveform level, confirm the  $> 20$  mm events show discharge/afterpulse signatures rather than clean muon clusters.

*Reproduce:* `extract_recofar.py` (categorise + join to `EventResult`) then `make_plots.py`; numbers in `recofar_meta.json`.  
Categorisation matches `09_efficiency_breakdown.py` (same alignment, active box, and 50-strip spark cut).